

# Labwork 1

Pham Do Ngoc Minh

May 2026

## 1 Implementation

- 2 functions  $f(x)$  and  $f'(x)$  to find the value of  $f(x)$  and  $f'(x)$
- For this lab, example is  $f(x) = x^2$  so  $f'(x) = 2x$
- Calculate the new value of  $x$  using the gradient descent formula:  $x = x - R * f'(x)$ . Loop that 10 times
- Print the value of  $x$  and  $f(x)$  each time

## 2 Analyze the learning rate

- When the learning rate  $r$  at 0.01, too small number makes the  $f(x)$  large, still around 70.
- When the learning rate  $r$  at 0.1, number around this makes the diversity look clearer.
- At 0.5, all 0.
- At 0.99, the number too large makes the  $f(x)$  same as 0.1, but  $x$  can be negative.